Khata Voice — Multilingual Voice Ledger for Kirana Stores

1-PAGE SOLUTION PITCH | The Great Indian AI Internship Challenge (Gnani.ai)

1. Problem Statement: Paper-Based Friction in Indian Retail

Over 12 million micro-retailers in Tier 2/3 India track credit (*udhar / khata*) in paper diaries. This creates severe business issues: manual recording causes lost entries during shop rush hours, shopkeepers lack structured debt recovery tracking, and existing accounting applications are complex, text-heavy, and lack localized Indic voice support.

2. Our Solution: Voice-First, Regional Credit Accounting

Khata Voice transforms store accounting into natural speech interactions across **Telugu, Hindi, and English**. Shopkeepers simply speak naturally: *'Ramu 500 rupees rice'* (■■■■ 500 ■■■■■■■ ■■■■■■■). The system transcribes the speech, validates customer identity, records the provision and credit balance, and immediately announces confirmation via a localized Smart Soundbox.

Real-World Flow: *[Speak Voice Note]* → ***Gnani Prisma v3.0 STT*** → *[Deterministic NLP]* → *[Confirm & Update Ledger]* → ***Smart Soundbox & WhatsApp Reminder***

3. Core Innovations & Gnani.ai Integration

- **Gnani Prisma v3.0 STT:** REST-integrated endpoint with customer vocabulary boosting to accurately recognize Indic regional vernaculars (te-IN, hi-IN, en-IN).
- **Zero-Hallucination Deterministic NLP:** Rule-based regex parser and fuzzy matching (difflib) mapping 55+ kirana provisions with vernacular aliases (bm → Rice, aata → Flour).
- **Smart Soundbox Feedback:** Instant multi-language audio verification using browser speech synthesis and Gnani Timbre, designed for noisy retail storefronts.
- **1-Click WhatsApp Payment Recovery:** One-tap generation of customized, polite reminder messages in the customer's native language with balance and item breakdown.
- **Multi-Tenant Isolation & Offline Mode:** Strict per-shopkeeper ledger sandboxing with PIN authentication, supplemented by an offline-ready keyboard entry mode.

4. Real-World Impact (Tier 2/3 Grounded)

Built and tested from Warangal, Telangana, Khata Voice cuts credit transaction time from **45 seconds down to < 5 seconds**. Polite, automated WhatsApp reminders accelerate debt recovery cycles by over 30%, safeguarding shopkeeper working capital.

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GitHub: github.com/charitha1203-create/khata-voice

Stack: Python, Streamlit, Gnani Prisma v3.0 STT, Pandas

Deliverables: 60s Demo Video, GitHub Repo, 1-Page Pitch